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# Mean or Mode: What the Loss Chooses When Colour Is Ambiguous

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## Abstract

Colourising a greyscale photograph has no single correct answer: a grey car is consistent with red, blue or white, and under a squared error the optimal prediction is then the mean of the plausible colours, which is close to grey. Two colourisers are trained on CIFAR-10 with everything but the loss held fixed, the second reading its predicted distribution at a temperature that interpolates between that distribution’s mean and its mode. Sweeping it traces a frontier on which accuracy and colour variety cannot both be maximised. At temperature one the classification network reproduces the squared error network pixel by pixel, 3.11 apart against the 12.49 separating either from the truth, and does so across independent seeds: the desaturation is a property of the question the loss asks, not of a model or a run.

## 1. Introduction

Given the lightness of a photograph, what colour was it? For many pixels the question has no answer: a greyscale car constrains its colour hardly at all, a greyscale frog constrains it a great deal. What exists is not a function from lightness to colour but a conditional distribution  $p(a, b \mid L)$ .

A network emits one colour per pixel, so something must collapse that distribution, and the loss decides what. Let  $\mu = \mathbb{E}[Y \mid X = x]$ . Adding and subtracting  $\mu$  inside the squared error and expanding gives

$$\mathbb{E}[\|\hat{y} - Y\|^2] = \|\hat{y} - \mu\|^2 + \text{Var}(Y \mid X = x), \quad (1)$$

since the cross term contains  $\mathbb{E}[\mu - Y] = 0$ . The second term is independent of the prediction, so the minimiser

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is  $\hat{y} = \mu$  (Bishop, 2006): the model produces not a plausible colour but the average of the plausible ones, and the average of red, blue and white is a barely tinted grey.

## 2. Related work

Posing colourisation as classification over quantised  $ab$  bins, and decoding with an annealed mean, is due to Zhang et al. (Zhang et al., 2016); Larsson et al. (Larsson et al., 2016) independently predicted per-pixel colour histograms. That a squared error averages ambiguous targets is likewise established (Mathieu et al., 2016). Neither is claimed as new. What is added is a controlled measurement: everything but the loss is held fixed, the frontier is traced by one knob rather than two tuned systems, and the networks are compared against each other pixel by pixel. Three departures from (Zhang et al., 2016) follow: their class rebalancing is a second mechanism for raising vividness and would confound the trade measured here, so the cross-entropy is left plain; their fixed 313-bin grid becomes the same 10-unit grid cut to the cells the data visits; and CIFAR-10 replaces ImageNet.

## 3. Method

Work is in CIE Lab, which separates the lightness the network is given from the chroma it must guess and in which euclidean distance approximates perceived colour difference. The network receives  $L$  and predicts  $(a, b)$  everywhere.

Architecture. Six blocks of  $3 \times 3$  convolution, batch normalisation (Ioffe & Szegedy, 2015) and ReLU at 64 channels, dilations 1, 1, 2, 4, 8, 1 (Yu & Koltun, 2016), then a  $1 \times 1$  convolution to the output. Dilation rather than striding keeps full resolution and gives a receptive field of 35 pixels, wider than the image, so residual uncertainty is ambiguity in the data and not myopia. Everything up to it is shared; the first model minimises the squared error on  $(a, b)$ , the second quantises the  $ab$  plane into 10-unit cells, keeps the 124 holding 99.9% of the training pixels, and minimises the cross-entropy

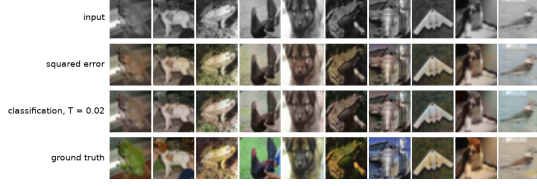


Figure 1. The two losses on the same test images, the classification model read at its mode. Both are muted against the truth, and muted in nearly the same way. One seed of each.

over them.

Readout. With bin centres  $c_k$  and probabilities  $p_k$ ,

$$\hat{y}(T) = \sum_k \frac{\exp(\log p_k/T)}{\sum_j \exp(\log p_j/T)} c_k, \quad (2)$$

the distribution’s mean at  $T = 1$ , which is what Equation (1) predicts the squared error model produces, and its mode as  $T \rightarrow 0$ .

Metrics and protocol. Colour error is the mean euclidean distance to the truth in  $ab$ ; vividness is mean chroma against the truth’s own. Since a model painting every pixel the same beige already scores 39.0% on it, variety, the mean distance from the model’s own average output and zero for any constant prediction, sits beside it. 3 seeds, 20 epochs, batch 128, AdamW (Loshchilov & Hutter, 2019) with a one-cycle schedule (Smith & Topin, 2019), in PyTorch (Paszke et al., 2019) on a laptop GPU. All numbers are on the 10 000 held-out test images of CIFAR-10 (Krizhevsky, 2009), brackets giving the range over seeds; labels are never shown to the colourisers. Code: <https://github.com/solazzo1911897-collab/colorization-mean-vs-mode>.

## 4. Results

Baselines and the cost of the mean. Grey everywhere gives 14.71 error; the average training colour gives 14.64 at 39.0% vividness and 0.0% variety. The squared error model reaches 12.44 [12.42, 12.47] at 62.0 [61.1, 63.1]% vividness and 53.8 [53.2, 54.4]% variety, beating both (Figure 1).

The frontier. Sweeping  $T$  (Figure 2) turns one prediction into a curve, and the squared error model sits on it: at  $T = 1$  the classification model gives 12.43 [12.41, 12.46] and 53.9 [53.5, 54.3]% variety against 12.44 [12.42, 12.47] and 53.8 [53.2, 54.4]%. Toward the mode variety rises and accuracy falls, from

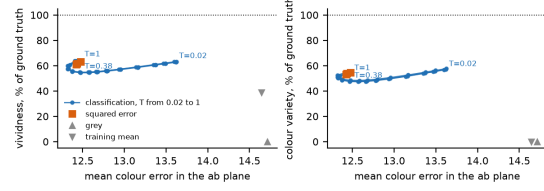


Figure 2. Accuracy against vividness (left) and variety (right) as the readout temperature is swept; all 3 seeds drawn. The constant baseline sits at 39.0% vividness and zero variety.

50.5 [50.2, 50.9]% at 12.32 [12.32, 12.33] ( $T = 0.65$ , lowest error) to 57.1 [56.7, 57.7]% at 13.61 [13.60, 13.63]. Neither end wins on both axes, and the squared error model cannot move along the curve at all. The range is narrow: even at the mode only 62.9 [62.8, 63.0]% of the truth’s vividness is kept against 62.0 [61.1, 63.1]%, the conditional distribution being concentrated enough here that its mean and its mode nearly coincide, so the readout alone cannot close the gap.

The two models are one model. Equation (1) predicts the squared error output should coincide with the expectation of the classification model’s distribution, and it does: 3.11 [3.07, 3.15] in  $ab$  units between them against 12.49 [12.47, 12.52] between either and the truth, a factor of 4.0; against that model’s mode it is 6.78 [6.64, 6.96] (Figure 3). The scale that makes that small is not the distance to the truth but the run-to-run floor: two squared error models differing only in seed are 2.73 [2.67, 2.77] apart, two classification models 2.45 [2.43, 2.48], so changing the loss moves the output about as far as changing the initialisation. Pairing across seeds changes nothing, 3.11 [3.08, 3.14].

A prediction that failed. The desaturation was expected to be worst where colour is least predictable, yet grouped by CIFAR-10 class the correlation with the squared error model’s vividness is +0.36 between instances and +0.02 over all pixels, and per pixel against the classification model’s own predictive entropy +0.69: no relation in the predicted direction, and one confound behind both (appendix).

Discussion. That the two models agree is what Equation (1) predicts, so this is a confirmation rather than a surprise. What it adds is that the agreement survives finite data and runs sharing nothing but the loss, locating the desaturation in the objective, not in any model. Vividness must therefore come from the readout or from reweighting rare colours, not from the loss.

## Statement on the use of AI

Anthropic’s Claude was used as an assistive tool throughout this project. It supported the refinement of the experimental methodology, assisted in extending and debugging the codebase, and aided in source verification and bibliographic assembly. It was also employed to polish, format, and synthesize drafts into the final report.

The Lab conversion was validated against published sRGB primaries, all citations were cross-checked at source, and all numerical values were derived directly from run logs. Failed predictions (e.g., Section 4) have been retained as observed.

## References

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## Appendix

**Limitations.** The resolution, at which colour ambiguity carries no texture component; a single dataset family; the absence of any perceptual evaluation, which for colourisation is what ultimately matters and requires human judgement rather than a distance in Lab; the temperature reported as lowest-error is read off the test sweep rather than a held-out split, so it is a description of the curve and not a tuned setting; and class rebalancing, excluded by design, whose contribution to vividness is the obvious next measurement.

The sweep doubles back. Both vividness and variety are U-shaped in  $T$  with a minimum near  $T = 0.3$ , where the softmax is flat enough to pull the readout toward the centroid of the bin grid but not flat enough to reach it, so the curve of Figure 2 folds rather than tracing a single monotone trade.

The grid is not the ceiling. Dropping the cells the training set barely visits is the obvious thing to suspect when a colouriser comes out desaturated. It is not the cause here. Quantising the ground truth itself to the 124 kept centres and scoring it as though it were a prediction gives 4.12 colour error at 109.4% vividness, slightly above the truth’s own rather than below it, since a near-grey pixel is moved out to a cell centre; 99.9% of test pixels fall inside a cell that was kept. The integer lookup that replaces the distance matrix when encoding a pixel agrees with the exact nearest centre 97.2% of the time and lands 0.003% further away when it does not.

**Ambiguity measured per pixel.** The failed prediction of Section 4 has a per-pixel form: the pixels where the classification model’s predicted distribution is widest should be the desaturated ones. Sorted into 10 equal-count buckets of predictive entropy (Figure 4) the correlation with vividness is +0.69 for the squared error model and +0.01 for the classification model, and the curve is not monotone in either. The measure carries

the same defect as the per-class one, and it is visible here: the true chroma of a bucket grows by a factor of 6.0 from the lowest entropy to the highest, so the denominator of the ratio moves with the quantity being tested. Whether desaturation tracks ambiguity is a question neither measurement answers.

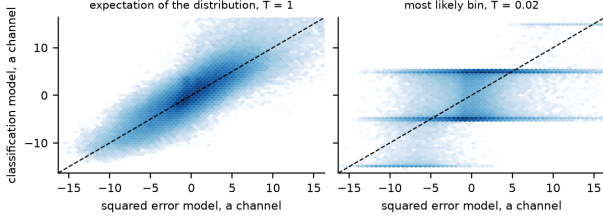


Figure 3. Squared error output against the classification readout, per pixel, on the  $a$  channel, for one seed of each. Left, the expectation of its distribution, which agrees along the diagonal; right, its most likely bin, where the correspondence is gone and the discrete bin centres band the plot. The distances quoted in the text average over all 3 seeds of each loss.

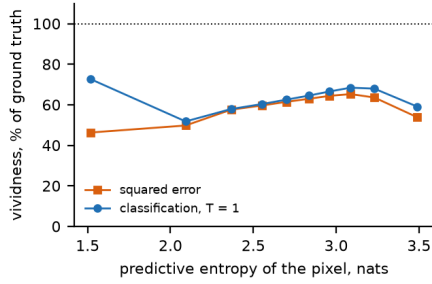


Figure 4. Vividness against the classification model's own predictive entropy, one point per equal-count bucket of test pixels. The predicted relation is a fall to the right; neither model shows one.

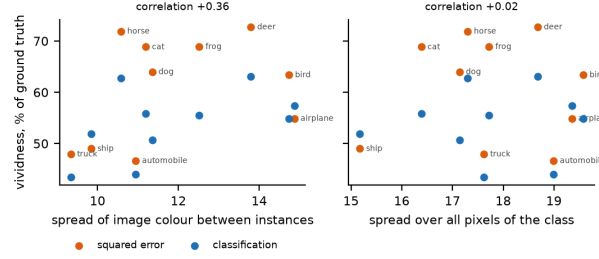


Figure 5. Vividness against two measures of how ambiguous the colour of a CIFAR-10 class is. Neither shows the predicted relation. Vividness is reported as a ratio to each class's own ground truth, so classes dominated by large saturated regions of sky or sea carry a high denominator whatever their ambiguity, and the measure conflates the two. Kept because the prediction was made before the measurement.

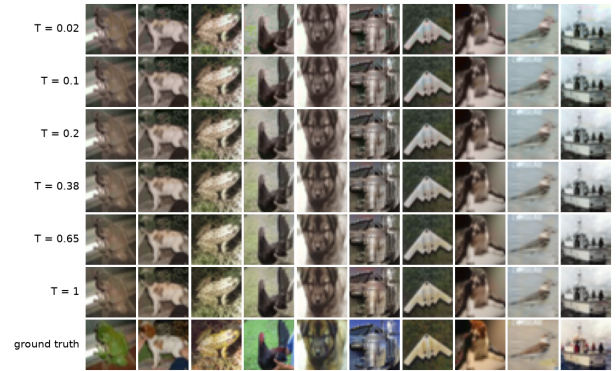


Figure 6. The same images decoded at increasing temperature, from the most likely colour to the mean of the distribution.